

- Explain what is meant by *autosomal linkage*.

A sweet pea plant that is heterozygous for both purple flowers and long pollen grains is crossed with a sweet pea plant with red flowers and round pollen grains. The results of this cross are shown in Table 8.1.

offspring phenotype	percentage of offspring
purple flowers, long pollen grains	44
red flowers, round pollen grains	44
purple flowers, round pollen grains	6
red flowers, long pollen grains	6

[4]

(c) Different alleles affecting flower colour in the sweet pea plant are the result of gene mutations.

State how a gene mutation may occur.

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.....[3]

[Total: 9]

Section B

Answer **one** question.